Simplification of the finite terms ($\Lambda \to 0$, $\vee \to \infty$): detailed note

Contents

1 Summary **1**

2 Conventions and building blocks **2**
2.1 General . . . 2
2.2 Kernels of one pair of vectors . . . 2
2.3 Colour objects . . . 2
2.4 Special functions and plus prescriptions . . . 3

3 Group I **3**
3.1 Row II . . . 3
3.2 Row III . . . 4
3.3 Row IV . . . 5
3.4 Row V, first part . . . 5
3.5 Row V, second part: $N_{2a} + N_{2b\,\mathrm{rest}}$ . . . 6

4 Group II **6**
4.1 Row I, first part . . . 6
4.2 Row I, second part ($2a$–$2f$) . . . 7
4.3 Row IV . . . 8

5 Group III **8**
5.1 Rows I and II . . . 8
5.2 Row III (first part) + Row V + Row VI . . . 8
5.3 Row III, second part . . . 9

6 The $K\log$ sector across the groups **10**

7 The negative logarithms and reality **10**

8 The simplified finite term **11**

9 Assumptions and points to check **11**

A Fourier transform of $K^{ij}\log(b^2/a^2)$ **12**

B Numerical checks **12**

1 Summary

The input is the list of finite terms in `Limiting_Cases.tex`: Group I (Rows II–V, with c.c.), Group II (Rows I, IV) and Group III (Rows I–III, V, VI, with c.c.). The results of this note are:

1. **Group I, Row II** (momentum space). The colour factor is antisymmetric under $x \leftrightarrow y$, which removes two of the three logarithmic structures of R^i. The remaining kernel has a single logarithm, Eq. (3.6), and is exactly the Fourier transform of the coordinate-space kernel $K^{ij}\log[(x-z)^2/(x-w)^2]$ that appears in the other rows, Eq. (3.11).

2. **Group I, Row III.** The logarithms of $(x-w)^2-(x-z)^2$ cancel completely, together with all poles at $(x-w)^2=(x-z)^2$. The 22 terms reduce to three structures, Eq. (3.24). One constant $i\sigma\pi$ survives; with the prefactor i it is a real contribution.

3. **The $K\log$ sector.** Six contributions (I.II, I.III, I.IV, II.I first part, III.I, III.II) have the same kernel. Their colour factors add up to commutators of ρ only, Eq. (6.6). For commuting ρ the whole sector cancels.

4. **Group II, Row I, second part** ($2a$–$2f$). The six pieces are a perfect square of one two-gluon amplitude, Eq. (4.11), with the closed form (4.12).

5. **Group III, Rows III (first part), V, VI** add up to one term whose colour factor, for commuting ρ, is a product of three LO factors, Eq. (5.3).

6. **Group I Row V and Group III Row III (second parts)** are written with the same three building blocks E, G, ϕ as the square of item 4; in Row V the two pieces N_{2a} and $N_{2b\,\mathrm{rest}}$ merge.

7. **Longitudinal integrals** that contain only a phase are done in closed form (Group I Row V first part, Group II Row IV, Group III).

8. **Reality.** Every group is real. The only complex kernel is the $i\sigma\pi$ of item 2.

The final list of what is left is in Section 8; assumptions and open points are in Section 9. Every algebraic statement was checked numerically (Appendix B).

2 Conventions and building blocks

2.1 General

$$n^i(r) \equiv \frac{r^i}{r^2}, \qquad M \cdot N \equiv M^{km} N^{km}, \qquad (M^T)^{km} \equiv M^{mk}. \tag{2.1}$$

Transverse vectors are written $a, b, c, \ldots$; they should not be confused with colour indices in U^{ab}. "+c.c." is the hermitian conjugate for operator-valued ρ. For Group II, $N_{1b} = (N_{1a})^*$, so $N_{1a} + N_{1b}$ is also of the form $X + \text{c.c.}$

2.2 Kernels of one pair of vectors

For a pair (a, b) with $c \equiv a - b$, $A \equiv a^2$, $B \equiv b^2$, $c^2 = A + B - 2a \cdot b$:

$$K^{ij}(a,b) \equiv \frac{1}{c^2}\left\{\frac{\delta^{ij}}{2} - \frac{a^i a^j}{A} - \frac{b^i b^j}{B} + \frac{2\,(a\cdot b)\,a^i b^j}{AB}\right\}, \tag{2.2}$$

$$S^{ij}(a,b) \equiv \frac{1}{c^2}\left\{\delta^{ij} - \frac{a^i a^j}{A} - \frac{b^i b^j}{B} + \frac{(A+B)\,a^i b^j}{AB}\right\} = K^{ij} + \frac{\delta^{ij}}{2c^2} + \frac{a^i b^j}{AB}, \tag{2.3}$$

$$T^{ij}(a,b) \equiv \frac{1}{c^2}\left\{\frac{(c\cdot a)}{A}\,a^i a^j + \frac{(c\cdot b)}{B}\,b^i b^j - \frac{(a\cdot b)\,(A-B)}{AB}\,a^i b^j\right\}, \tag{2.4}$$

and

$$E^{km}(a,b) \equiv a^k\left[\frac{c^m}{c^2} + \frac{b^m}{2B}\right], \qquad G^{km}(a,b) \equiv \left[\frac{c^k}{c^2} - \frac{a^k}{2A}\right] b^m, \qquad \phi(a,b) \equiv \frac{B-A}{2\,c^2}, \tag{2.5}$$

with traces $e \equiv E^{kk} = \frac{a\cdot c}{c^2} + \frac{a\cdot b}{2B}$ and $g \equiv G^{kk} = \frac{b\cdot c}{c^2} - \frac{a\cdot b}{2A}$. Identities used repeatedly:

$$c\cdot a = A - a\cdot b, \qquad c\cdot b = a\cdot b - B, \qquad A - B = c\cdot(a+b). \tag{2.6}$$

Unless stated otherwise the pair is

$$a = x - w, \qquad b = x - z, \qquad c = z - w. \tag{2.7}$$

2.3 Colour objects

$$\begin{aligned}
\mathcal{D}_0^a &\equiv \big[U^{ab'}(x') - U^{ab'}(w')\big]\rho^{b'}(x'), & \mathcal{D}^c(u,v) &\equiv \big[U^{ce}(u) - U^{ce}(v)\big]\rho^e(v),\\
R^{ab} &\equiv f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x), & V^{be} &\equiv U^{cb}(z)\,U^{ce}(y),\\
\mathcal{S}^a &\equiv f^{hbc}\,U^{ah}(w)\{\rho^b(x),\rho^c(y)\} - f^{abc}\,U^{bd}(x)\,U^{ce}(y)\{\rho^d(x),\rho^e(y)\}.
\end{aligned} \tag{2.8}$$

R'^{ab} is R^{ab} with $(w,x) \to (w',x')$ and the same z. $\mathcal{S}^a$ is the colour bracket of Group I Row II.

2.4 Special functions and plus prescriptions

$$E_1(z)=\int_1^\infty \frac{dt}{t}\,e^{-zt},\qquad \mathrm{Ein}(z)=\int_0^1 \frac{dt}{t}\,(1-e^{-zt})=\gamma_E+\log z+E_1(z). \tag{2.9}$$

For real $s>0$: $E_1(is)=-\mathrm{Ci}(s)+i\,[\mathrm{Si}(s)-\frac{\pi}{2}]$, $\mathrm{Ein}(is)=\gamma_E+\log s-\mathrm{Ci}(s)+i\,\mathrm{Si}(s)$. The plus prescriptions are read as

$$\left[\frac{1}{\xi}\right]_+ f\equiv\frac{f(\xi)-f(0)}{\xi}\quad(0<\xi<1),\qquad \left[\frac{1}{\eta(1+\eta)}\right]_+ f\equiv\frac{f(\eta)-f(0)}{\eta(1+\eta)}\quad(0<\eta<\infty). \tag{2.10}$$

3 Group I

3.1 Row II

Step 1: rename. The integration momentum of Row II is called k in the file, the same symbol as the external momentum. I call it q, and I assume the integral over x' is implied:

$$\frac{d^3N_{\mathrm{I.II}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w',x'}e^{-ik\cdot(w'-w)}\int d^2p\,d^2q\int_{x,y}e^{iq\cdot(w-y)}\,e^{ip\cdot(y-x)}\;n^i(x'-w')\,\frac{R^i(q,p)}{p^2q^2}\;\mathcal{D}_0^a\,\mathcal{S}^a+\text{c.c.} \tag{3.1}$$

$$R^i(q,p)=-2q^i\log\frac{p^2}{q^2}+\frac{(2p^2-2\,q\cdot p+q^2)\,q^i+q^2p^i}{(q-p)^2}\;\log\frac{q^2}{(q-p)^2}. \tag{3.2}$$

Step 2: the colour factor is antisymmetric under $x\leftrightarrow y$. Exchanging $x\leftrightarrow y$ in $\mathcal{S}^a$ and relabelling $b\leftrightarrow c$ ($d\leftrightarrow e$) gives $f^{hcb}=-f^{hbc}$ while the anticommutators are symmetric, hence

$$\mathcal{S}^a\big|_{x\leftrightarrow y}=-\,\mathcal{S}^a \tag{3.3}$$

exactly, also for non-commuting ρ. Nothing else in the integrand depends on x,y except the phases, so only the part of the kernel that is antisymmetric under $x\leftrightarrow y$ contributes.

Step 3: action of $x\leftrightarrow y$ on the momenta. With $l\equiv p-q$,

$$e^{iq\cdot(w-y)}e^{ip\cdot(y-x)}\;\xrightarrow{x\leftrightarrow y}\;e^{iq\cdot(w-x)}e^{ip\cdot(x-y)}=e^{iq\cdot(w-y)}\,e^{i(q-p)\cdot(y-x)}, \tag{3.4}$$

i.e. $p\to q-p$ at fixed q, which is $p^2\leftrightarrow l^2$. A term of the kernel that is symmetric under $p^2\leftrightarrow l^2$ at fixed q drops out.

Step 4: split the kernel. Using $2p^2-2q\cdot p+q^2=p^2+l^2$ and $p=q+l$,

$$\frac{R^i}{p^2q^2}=-\frac{2q^i}{p^2q^2}\log\frac{p^2}{q^2}+\left[\frac{q^i}{q^2l^2}+\frac{q^i}{p^2q^2}+\frac{p^i}{p^2l^2}\right]\log\frac{q^2}{l^2}=\mathcal{R}^i(q,p)+2\,\mathcal{E}^i(q,p), \tag{3.5}$$

$$\boxed{\mathcal{R}^i(q,p)\equiv\log\frac{q^2}{(p-q)^2}\left[\frac{p^i}{p^2\,(p-q)^2}-\frac{q^i}{q^2\,(p-q)^2}+\frac{q^i}{p^2q^2}\right]},\qquad \mathcal{E}^i\equiv\frac{q^i}{q^2}\left[\frac{1}{l^2}\log\frac{q^2}{l^2}-\frac{1}{p^2}\log\frac{p^2}{q^2}\right]. \tag{3.6}$$

$\mathcal{E}^i$ is symmetric under $p^2\leftrightarrow l^2$ and drops out by Step 3. It contains the whole term $-2q^i\log(p^2/q^2)$ and the non-integrable $l\to0$ behaviour $\sim\log(l^2)/l^2$ of the original kernel; $\mathcal{R}^i$ is only $\mathcal{O}(1/|l|)$ at $l\to0$ because $p^i/p^2-q^i/q^2=\mathcal{O}(l)$. So

$$\frac{d^3N_{\mathrm{I.II}}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w',x'}e^{-ik\cdot(w'-w)}\int d^2p\,d^2q\int_{x,y}e^{iq\cdot(w-y)}\,e^{ip\cdot(y-x)}\;n^i(x'-w')\,\mathcal{R}^i(q,p)\;\mathcal{D}_0^a\,\mathcal{S}^a+\text{c.c.} \tag{3.7}$$

Step 5: coordinate space. The Fourier transform of the kernel $K^{ij}\log(b^2/a^2)$ is computed in Appendix A:

$$l^j\int d^2a\,d^2b\;e^{iq\cdot a+il\cdot b}\,K^{ij}(a,b)\,\log\frac{b^2}{a^2}=-2\pi^2\,l^2\,\mathcal{R}^i(q,q+l). \tag{3.8}$$

Define $H^i \equiv \int_z n^j(y-z)\,K^{ij}(a,b)\log(B/A)$ with $a=x-w$, $b=x-z$, and $\rho\equiv y-x$ so that $y-z=\rho+b$. Its transform is

$$\int d^2a\,d^2\rho\,e^{iq\cdot a-il\cdot\rho}\,H^i=\underbrace{\int d^2r\,e^{-il\cdot r}\,n^j(r)}_{-2\pi i\,l^j/l^2}\int d^2a\,d^2b\,e^{iq\cdot a+il\cdot b}K^{ij}\log\frac{b^2}{a^2}=4\pi^3 i\,\mathcal{R}^i(q,q+l). \tag{3.9}$$

Since $e^{iq\cdot(w-y)}e^{ip\cdot(y-x)}=e^{-iq\cdot a}e^{il\cdot\rho}$, inverting gives

$$\int d^2p\,d^2q\,e^{iq\cdot(w-y)}e^{ip\cdot(y-x)}\,\mathcal{R}^i(q,p)=\frac{(2\pi)^4}{4\pi^3 i}\,H^i=-4\pi i\,H^i, \tag{3.10}$$

and (3.7) becomes

$$\boxed{\frac{d^3N_{\rm I.II}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\int_{w,w',x,y,z,x'}e^{-ik\cdot(w'-w)}\;n^i(x'-w')\,n^j(y-z)\,K^{ij}\;\log\frac{(x-z)^2}{(x-w)^2}\;\;\tfrac{1}{2}\,\mathcal{D}_0^a\,\mathcal{S}^a+\text{c.c.}} \tag{3.11}$$

Row II is therefore one more term of the $K\log$ type (Section 6).

3.2 Row III

All terms of N_1, N_2, N_{3a}, N_{3b} carry the prefactor $ig^4/8\pi^5k^+$ (some with an extra $\frac{1}{2}$) and the same colour factor

$$\mathcal{W}_{\rm III}\equiv\mathcal{D}_0^a\,f^{abd}\,U^{de}(x)\big[U^{bc}(z)-U^{bc}(y)\big]\rho^e(x)\,\rho^c(y), \tag{3.12}$$

which does not depend on w. With $u^i\equiv n^i(x'-w')$, $v^j\equiv n^j(y-z)$ write

$$\frac{d^3N_{\rm I.III}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\int_{w,w',x,y,z,x'}e^{-ik\cdot(w'-w)}\;u^i\,v^j\;\mathcal{R}^{ij}_{\rm III}\,\mathcal{W}_{\rm III}\;+\;\text{c.c.} \tag{3.13}$$

Step 1: the four pieces. Reading off the file (the overall sign of N_2 is $-$):

$$\begin{aligned}
\mathcal{R}_1^{ij}&=\frac{\delta^{ij}}{2}\,\frac{A+B}{(B-A)^2}\left\{\log\frac{B}{B-A}+\log\frac{A-B}{A}\right\}-\frac{a^i\,b^j}{A\,B}\,\log\frac{-B}{A},\\
\mathcal{R}_2^{ij}&=-\,M^{ij}\,\log\frac{-B}{A-B},\\
\mathcal{R}_{3a}^{ij}&=-\,M^{ij}\,\log\frac{A-B}{A}+\mathcal{A}_0\left[\frac{c^ia^j}{c^2}+\frac{b^ic^j}{c^2}-\frac{(a\cdot b)\,c^ib^j}{c^2\,B}\right],\\
\mathcal{R}_{3b}^{ij}&=-\,\mathcal{A}_0\,\frac{(a\cdot b)\,a^ic^j}{c^2\,A}-\frac{1}{2}\,\log\frac{c^2}{A}\left[\frac{c^ib^j}{c^2B}-\frac{a^ic^j}{c^2A}+\frac{\delta^{ij}}{c^2}\right],
\end{aligned} \tag{3.14}$$

where the same tensor appears in N_2 (six terms) and in the six logarithmic terms of N_{3a} (with opposite sign):

$$M^{ij}\equiv\frac{1}{c^2}\left\{\frac{c^ia^j+b^ic^j}{A-B}-\frac{1}{A-B}\left[\frac{B}{A}\,a^ic^j+\frac{A}{B}\,c^ib^j\right]+\delta^{ij}\,\frac{B\,(c\cdot a)-A\,(c\cdot b)}{(A-B)^2}\right\}. \tag{3.15}$$

Step 2: the logarithms of $(A-B)$ cancel. With $L_a\equiv\log\frac{B}{B-A}=\log\frac{-B}{A-B}$, $L_b\equiv\log\frac{A-B}{A}$, $L_c\equiv\log\frac{-B}{A}$: for $A>B$ the argument of L_a is negative and that of L_b positive, for $A<B$ the reverse, and in both regions

$$L_a+L_b=\log\left[\frac{B}{B-A}\cdot\frac{A-B}{A}\right]=L_c=\log\frac{B}{A}+i\,\sigma\,\pi,\qquad\sigma=\pm1. \tag{3.16}$$

$\log|A-B|$ drops out. The remnant of the negative arguments is the constant $i\sigma\pi$ with the same sign in both regions ($\sigma=+1$ for the principal branch, $\sigma=-1$ for the opposite $i\epsilon$). Then $\mathcal{R}_2+\mathcal{R}_{3a}|_{\log}=-M^{ij}(L_a+L_b)=-M^{ij}L_c$ and

$$\mathcal{R}^{ij}_{\rm III}\Big|_{L_c}=\log\frac{-B}{A}\left\{\frac{\delta^{ij}}{2}\,\frac{A+B}{(A-B)^2}-\frac{a^ib^j}{AB}-M^{ij}\right\}. \tag{3.17}$$

Step 3: the poles at $A = B$ cancel. For the δ^{ij} part, (2.6) gives $B\,(c\cdot a) - A\,(c\cdot b) = 2AB - (a\cdot b)(A+B)$ and

$$\frac{A+B}{2(A-B)^2} - \frac{2AB-(a\cdot b)(A+B)}{c^2(A-B)^2} = \frac{\frac{1}{2}(A+B)(A+B-2a\cdot b) - 2AB + (a\cdot b)(A+B)}{c^2(A-B)^2} = \frac{1}{2c^2}. \tag{3.18}$$

For the rest, with $c = a - b$: $c^i a^j + b^i c^j = a^i a^j - b^i b^j$ and $\frac{B}{A}a^i c^j + \frac{A}{B}c^i b^j = \frac{B}{A}a^i a^j - \frac{A}{B}b^i b^j + a^i b^j \frac{A^2-B^2}{AB}$, so

$$M^{ij}\Big|_{\text{non}-\delta} = \frac{1}{c^2}\left\{\frac{a^i a^j}{A} + \frac{b^i b^j}{B} - \frac{(A+B)\,a^i b^j}{AB}\right\}. \tag{3.19}$$

Adding $-a^i b^j/(AB) = -c^2 a^i b^j/(c^2 AB)$ and using $A + B - c^2 = 2a\cdot b$,

$$\mathcal{R}^{ij}_{\rm III}\Big|_{L_c} = K^{ij}\,\log\frac{-B}{A}, \tag{3.20}$$

with K^{ij} of (2.2). Writing $c = a - b$ in the bracket of Group I Row IV (the same bracket appears in Group II Row I ($1a$) and Group III Rows I, II) shows it is the same tensor:

$$\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2} - \frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2} - \frac{(x-w)^i(x-z)^j}{(x-w)^2(x-z)^2} + \frac{\delta^{ij}}{2(z-w)^2} = K^{ij}. \tag{3.21}$$

Step 4: the $\mathcal{A}_0$ terms.

$$\mathcal{R}^{ij}_{\rm III}\Big|_{\mathcal{A}_0} = \frac{\mathcal{A}_0}{c^2}\left\{c^i a^j + b^i c^j - (a\cdot b)\left[\frac{c^i b^j}{B} + \frac{a^i c^j}{A}\right]\right\} = \mathcal{A}_0\,T^{ij}, \tag{3.22}$$

with T^{ij} of (2.4). With $\Delta \equiv \sqrt{AB - (a\cdot b)^2}$ (twice the area of the triangle x, z, w), $\arctan[(a\cdot b)/\Delta] = \frac{\pi}{2} - \theta_x$ and $\arctan[(B - a\cdot b)/\Delta] = \frac{\pi}{2} - \theta_z$ where θ_x, θ_z are the interior angles at x and z. Hence

$$\mathcal{A}_0 = \frac{\pi - \theta_x - \theta_z}{\Delta} = \frac{\theta_w}{\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2}}, \qquad \cos\theta_w = \frac{(x-w)\cdot(z-w)}{|x-w|\,|z-w|}, \tag{3.23}$$

the interior angle at w divided by twice the area: real, no logarithm.

Step 5: the $\log[(z-w)^2/(x-w)^2]$ terms have a positive argument and give $-\frac{1}{2}\log\frac{c^2}{A}\,S^{ij}$ with S^{ij} of (2.3).

Step 6: result.

$$\boxed{\mathcal{R}^{ij}_{\rm III} = K^{ij}\left\{\log\frac{(x-z)^2}{(x-w)^2} + i\,\sigma\,\pi\right\} - \frac{1}{2}\,S^{ij}\,\log\frac{(z-w)^2}{(x-w)^2} + \mathcal{A}_0\,T^{ij}} \tag{3.24}$$

3.3 Row IV

Using $U^{bd}(z)U^{be}(y) = V^{de}$ the colour bracket of Row IV is $V^{be}\rho^e(y)\,R^{ab}$ (first term: $V^{de}\rho^e(y) f^{cdc'}U^{ac}(w)\rho^{c'}(x)$; second term: $U^{cb}(z)U^{gb}(z) = \delta^{cg}$). So

$$\frac{d^3N_{\rm I.IV}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\,u^i v^j K^{ij}\log\frac{(x-z)^2}{(x-w)^2}\,\mathcal{D}^a_0\,V^{be}\rho^e(y)\,R^{ab} + \text{c.c.} \tag{3.25}$$

3.4 Row V, first part

With $\sigma \equiv k\cdot(z-x)$ and (2.10):

$$\int_0^1 d\xi\,e^{-i\xi\sigma} = \frac{1-e^{-i\sigma}}{i\sigma} \equiv h_0(\sigma), \qquad \int_0^1 d\xi\left[\frac{1}{\xi}\right]_+ e^{\mp i\xi\sigma} = -\text{Ein}(\pm i\sigma). \tag{3.26}$$

The prefactors are $\frac{1}{k^{+2}}\int_0^{k^+}\frac{dp^+}{2\pi} = \frac{1}{2\pi k^+}\int_0^1 d\xi$ and $\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}[\frac{1}{q^+}]_+ = \frac{1}{2\pi k^+}\int_0^1 d\zeta\,[\frac{1}{\zeta}]_+$. Then

$$\begin{aligned}\frac{d^3N_{\rm I.V(1)}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\int_{x,z,y,x',y',w'}\Big\{&e^{-ik\cdot(w'-z)}\Big[h_0(\sigma)\;n(x'-w')\cdot n(z-x)\;\;n(y-z)\cdot n(y'-x)\\ &+\text{Ein}(i\sigma)\;n(y'-x)\cdot n(z-x)\;\;n(x'-w')\cdot n(y-z)\Big]\\ &+e^{-ik\cdot(w'-x)}\,\text{Ein}(-i\sigma)\;n(x'-w')\cdot n(y'-x)\;\;n(z-x)\cdot n(y-z)\Big\}\,\mathcal{D}^a_0\\ &\times\Big[f^{adc}\,U^{db}(x)U^{ce}(z)\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)U^{be}(y')\{\rho^d(y),\rho^e(y')\}\Big] + \text{c.c.}\end{aligned} \tag{3.27}$$

3.5 Row V, second part: $N_{2a} + N_{2b\,\mathrm{rest}}$

Use the blocks (2.5) at $\bar a \equiv y-x$, $\bar b \equiv y-z$, $\bar c = \bar a - \bar b = z-x$ (bars on E, G, ϕ, e, g), and $\xi = p^+/k^+$.

Step 1: the bracket of $N_{2b\,\mathrm{rest}}$. Term by term ($x - z = -\bar c$):

$$
\begin{aligned}
\frac{\delta^{im} d_\perp}{2(x-z)^2}\big[\bar a^2 - \bar b^2\big]\xi(1-\xi) &= -d_\perp\,\xi(1-\xi)\,\bar\phi\,\delta^{im},\\
\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2} &= -\bar E^{im},\\
\frac{(y-z)\cdot(x-z)}{(x-z)^2} + \frac{(y-x)\cdot(y-z)}{2(y-x)^2} - \frac{\bar a^2-\bar b^2}{2(x-z)^2} &= -\bar g + \bar\phi,\\
\frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2} &= -\bar G^{mi},\\
\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2(y-z)^2} - \frac{\bar a^2-\bar b^2}{2(x-z)^2} &= -\bar e + \bar\phi,\\
\frac{(y-x)^m(x-z)^i}{(x-z)^2} - \frac{(y-x)^m(y-z)^i}{2(y-z)^2} + \frac{(y-z)^m(x-z)^i}{(x-z)^2} + \frac{(y-x)^i(y-z)^m}{2(y-x)^2} &= -\bar E^{mi} - \bar G^{im}.
\end{aligned}
\tag{3.28}
$$

The bracket is therefore

$$
\begin{aligned}
\mathfrak{B}_V^{im}(\xi) = \delta^{im}\Big[\bar\phi\,\big(1 - d_\perp\,\xi(1-\xi)\big) - \xi\,\bar e - (1-\xi)\,\bar g\Big]\\
+ \left(\left[\frac{1}{1-\xi}\right]_+ - 1\right)\bar E^{im} + \left(\left[\frac{1}{\xi}\right]_+ - 1\right)\bar G^{mi} + \bar E^{mi} + \bar G^{im}.
\end{aligned}
\tag{3.29}
$$

Step 2: N_{2a} has the same bracket. Its first integral is (3.29) without the $\bar E^{im}$ term. In its second integral put $q^+ = k^+ - p^+$: the phase $e^{-ik\cdot(w'-x)}e^{-ik\cdot(x-z)q^+/k^+}$ becomes $e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)p^+/k^+}$, the denominator becomes $p^+\bar a^2 + (k^+-p^+)\bar b^2$, and $\int_{k^+}^0 dq^+ = -\int_0^{k^+} dq^+$ (limits as written in the file) gives $+([\frac{1}{1-\xi}]_+ - 1)\bar E^{im}$. So N_{2a} and $N_{2b\,\mathrm{rest}}$ share the kernel $\mathfrak{B}_V^{im}/[\xi\bar a^2 + (1-\xi)\bar b^2]$.

Step 3: add the colour factors. $\big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_c U^{af}(x)\big] + N_c\big[U^{af}(y) - U^{af}(x)\big]$: the $U(x)$ terms cancel.

$$
\begin{aligned}
\frac{d^3N_{\mathrm{I.V}(2)}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_0^1 d\xi \int_{x,y,z,x',w'} \Bigg\{ e^{-ik\cdot(w'-z)}\,e^{-i\xi\,k\cdot(z-x)}\;\frac{n^i(x'-w')\,\mathfrak{B}_V^{im}(\xi)\,n^m(z-x)}{\xi\,(y-x)^2 + (1-\xi)\,(y-z)^2}\\
\times\big[U^{ab'}(x') - U^{ab'}(w')\big]\Big[N_c\,U^{af}(y) - f^{adc}f^{fbe}\,U^{db}(x)\,U^{ce}(z)\Big]\\
- e^{-ik\cdot(w'-x)}\,\big[\xi(1-\xi) - 2\big]\;\frac{n(x'-w')\cdot(y-x)\;\; n(y-z)\cdot n(y-x)}{(x-z)^2}\\
\times N_c\big[U^{ab'}(x') - U^{ab'}(w')\big]\big[U^{af}(y) - U^{af}(x)\big]\Bigg\}\rho^{b'}(x')\rho^f(y) + \text{c.c.}
\end{aligned}
\tag{3.30}
$$

The colour factor of the first term is that of a gluon at y splitting into gluons at x and z. At $z \to x$ it becomes $N_c[U(y) - U(x)]$, because $f^{adc}f^{fbe}U^{db}(x)U^{ce}(x) = N_c U^{af}(x)$, which is the colour factor of the subtraction, and $\xi(1-\xi) - 2$ is the plus-prescription form of $\frac{\xi}{1-\xi} + \frac{1-\xi}{\xi} + \xi(1-\xi)$.

4 Group II

4.1 Row I, first part

With $u^i = n^i(x'-w')$, $v^j = n^j(y-z)$,

$$
\begin{aligned}
\frac{d^3N_{1a}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\,u^i v^j K^{ij}\,\log\frac{(x-z)^2}{(x-w)^2}\\
\times\Big[U^{ac'}(w')\{\rho^{c'}(x'),\rho^b(y)\} - U^{ad'}(x')\,V^{be'}\{\rho^{d'}(x'),\rho^{e'}(y)\}\Big]R^{ab}, \qquad N_{1b} = (N_{1a})^*.
\end{aligned}
\tag{4.1}
$$

4.2 Row I, second part ($2a$–$2f$)

Step 1: notation and colour.

$$a = x - w, \quad b = x - z, \quad c = z - w, \qquad a' = x' - w', \quad b' = x' - z, \quad c' = z - w' = a' - b',$$
$$\ell \equiv \log\frac{B}{A}, \qquad \ell' \equiv \log\frac{B'}{A'}, \qquad D \equiv A'B - AB'. \tag{4.2}$$

A prime on E, G, ϕ, e, g means evaluation at (a', b'). Using $U^{c'b}(z)U^{cb}(z) = \delta^{c'c}$ and $f^{acd'}f^{acd} = N_c\delta^{d'd}$, the four-term colour bracket common to all six pieces is

$$f^{c'bd'}f^{cbd}\,U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^{d}(x) - \cdots + N_c\,U^{de'}(x')U^{de}(x)\rho^{e'}(x')\rho^{e}(x) = R'^{ab}R^{ab}. \tag{4.3}$$

So

$$\frac{d^3N_{2a+\cdots+2f}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik\cdot(w'-w)}\;\mathcal{K}_{\rm II}\;R'^{ab}R^{ab}. \tag{4.4}$$

Step 2: the brackets are products of the blocks.

$$P \equiv [\dots]_{2a} = E\cdot G' + G\cdot E', \qquad X \equiv [\dots]_{2b} = E\cdot E', \qquad Y \equiv [\dots]_{2c} = G\cdot G',$$
$$Z \equiv [\dots]_{2d} = \phi'\,e + \phi\,e', \qquad V_e \equiv [\dots]_{2e} = \phi'\,g + \phi\,g', \qquad \frac{d_\perp}{4}\frac{(B-A)(B'-A')}{c^2c'^2} = d_\perp\,\phi\,\phi'. \tag{4.5}$$

For example $E\cdot E' = (a\cdot a')\big[\frac{c}{c^2} + \frac{b}{2B}\big]\cdot\big[\frac{c'}{c'^2} + \frac{b'}{2B'}\big]$ is the bracket of $2b$; $E\cdot G'$ gives terms 1–4 of $2a$ and $G\cdot E'$ terms 5–8.

Step 3: coefficients of the logarithms (in units of the prefactor of (4.4)):

$$2a:\ \frac{P}{D}\,(\ell-\ell'), \qquad 2d:\ \frac{Z}{D}\left[\frac{B}{B-A}\,\ell - \frac{B'}{B'-A'}\,\ell'\right],$$
$$2b:\ \frac{X}{D}\left[-\frac{B}{A}\,\ell + \frac{B'}{A'}\,\ell'\right], \qquad 2e:\ \frac{V_e}{D}\left[-\frac{A}{B-A}\,\ell + \frac{A'}{B'-A'}\,\ell'\right],$$
$$2c:\ \frac{Y}{D}\left[-\frac{A}{B}\,\ell + \frac{A'}{B'}\,\ell'\right], \qquad 2f:\ -d_\perp\phi\phi'\left\{\frac{1}{D}\left[\frac{AB\,\ell}{(B-A)^2} - \frac{A'B'\,\ell'}{(B'-A')^2}\right] + \frac{1}{(B-A)(B'-A')}\right\}. \tag{4.6}$$

Step 4: each combination is one integral. With $\Delta(\eta) \equiv (\eta B + A)(\eta B' + A')$, by partial fractions:

$$2a:\quad \int_0^\infty \frac{d\eta}{\Delta} = \frac{\ell-\ell'}{D},$$
$$2b:\quad \int_0^\infty \frac{d\eta}{\eta}\left[\frac{1}{\Delta} - \frac{\theta(1-\eta)}{AA'}\right] = \frac{1}{D}\left[-\frac{B}{A}\ell + \frac{B'}{A'}\ell'\right],$$
$$2c:\quad \int_0^\infty d\eta\left[\frac{\eta}{\Delta} - \frac{\theta(\eta-1)}{BB'\,\eta}\right] = \frac{1}{D}\left[-\frac{A}{B}\ell + \frac{A'}{B'}\ell'\right],$$
$$2d:\quad \int_0^\infty \frac{d\eta}{(1+\eta)\,\Delta} = \frac{1}{D}\left[\frac{B\,\ell}{B-A} - \frac{B'\,\ell'}{B'-A'}\right],$$
$$2e:\quad \int_0^\infty \frac{\eta\,d\eta}{(1+\eta)\,\Delta} = \frac{1}{D}\left[-\frac{A\,\ell}{B-A} + \frac{A'\,\ell'}{B'-A'}\right],$$
$$2f:\quad \int_0^\infty \frac{\eta\,d\eta}{(1+\eta)^2\,\Delta} = -\frac{1}{D}\left[\frac{AB\,\ell}{(B-A)^2} - \frac{A'B'\,\ell'}{(B'-A')^2}\right] - \frac{1}{(B-A)(B'-A')}. \tag{4.7}$$

The two subtractions are the endpoint subtractions at $\eta \to 0$ (the $\Lambda \to 0$ side) and $\eta \to \infty$ (the $\vee \to \infty$ side).

Step 5: perfect square. Adding the six pieces,

$$\mathcal{K}_{\rm II} = \int_0^\infty d\eta\,\frac{\mathcal{N}(\eta)}{\Delta(\eta)}\bigg|_{\rm subtr.}, \qquad \mathcal{N}(\eta) = \frac{X}{\eta} + P + \eta\,Y + \frac{Z+\eta V_e}{1+\eta} + \frac{d_\perp\,\phi\phi'\,\eta}{(1+\eta)^2}, \tag{4.8}$$

and by Step 2 the numerator factorises ($\delta^{kk} = d_\perp$):

$$\mathcal{N}(\eta) = \Phi^{km}(a,b;\eta)\,\Phi^{km}(a',b';\eta), \qquad \Phi^{km}(a,b;\eta) \equiv \frac{1}{\sqrt{\eta}}\,E^{km} + \sqrt{\eta}\,G^{km} + \frac{\sqrt{\eta}}{1+\eta}\,\phi\,\delta^{km}. \tag{4.9}$$

In the original variables

$$\Phi^{km} = \frac{(x-w)^k}{\sqrt{\eta}}\left[\frac{(z-w)^m}{(z-w)^2} + \frac{(x-z)^m}{2(x-z)^2}\right] + \sqrt{\eta}\left[\frac{(z-w)^k}{(z-w)^2} - \frac{(x-w)^k}{2(x-w)^2}\right](x-z)^m + \frac{\sqrt{\eta}}{1+\eta}\,\frac{(x-z)^2-(x-w)^2}{2\,(z-w)^2}\,\delta^{km}. \tag{4.10}$$

Hence

$$\boxed{\frac{d^3N_{2a+\cdots+2f}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik\cdot(w'-w)}\int_0^\infty d\eta\,\left[\frac{\Phi'^{km}\,R'^{ab}}{\eta B'+A'}\right]\left[\frac{\Phi^{km}\,R^{ab}}{\eta B+A}\right]\Bigg|_{\text{subtr.}}} \tag{4.11}$$

"subtr.": in the $E\cdot E'/\eta$ term $1/\Delta \to 1/\Delta - \theta(1-\eta)/(AA')$; in the $\eta\,G\cdot G'$ term $\eta/\Delta \to \eta/\Delta - \theta(\eta-1)/(BB'\eta)$.

Closed form. The coefficient of each logarithm is the residue of the integrand:

$$\boxed{\mathcal{K}_{\rm II} = \frac{1}{D}\left\{\mathcal{N}\big(-\tfrac{A}{B}\big)\log\frac{B}{A} - \mathcal{N}\Big(-\tfrac{A'}{B'}\Big)\log\frac{B'}{A'}\right\} - \frac{d_\perp}{4\,c^2c'^2}} \tag{4.12}$$

$$\mathcal{N}\big(-\tfrac{A}{B}\big) = P - \frac{B}{A}X - \frac{A}{B}Y + \frac{BZ-AV_e}{B-A} - \frac{d_\perp\,\phi\phi'\,AB}{(B-A)^2}, \tag{4.13}$$

and $\mathcal{N}(-A'/B')$ is the same with $A\to A'$, $B\to B'$ in the explicit coefficients.

Consequences.

- At $D=0$ one has $A/B = A'/B'$: residues and logarithms coincide and the $1/D$ cancels. At $B=A$ the apparent poles are compensated by $\ell \simeq (B-A)/A$ and $\phi\propto(B-A)$.
- $\mathcal{K}_{\rm II}$ is real and symmetric under (primed $\leftrightarrow$ unprimed). For commuting ρ the unsubtracted integrand is $|\mathcal{M}|^2$ with $\mathcal{M}^{km,ab}(z,\eta) = \int_{w,x} e^{ik\cdot w}\Phi^{km}R^{ab}/(\eta B+A)$.
- All logarithms have positive arguments.

4.3 Row IV

With $r\equiv y'-y$, $s\equiv k\cdot r$, $t=1/\xi$:

$$\mathcal{H}(s) \equiv \int_0^1 d\xi\,\frac{1-\xi}{\xi}\,e^{-is/\xi} = \int_1^\infty dt\left(\frac{1}{t}-\frac{1}{t^2}\right)e^{-ist} = (1+is)\,E_1(is) - e^{-is}. \tag{4.14}$$

For $s>0$: $\mathrm{Re}\,\mathcal{H} = -\mathrm{Ci}(s) - s[\mathrm{Si}(s)-\frac{\pi}{2}] - \cos s$, $\mathrm{Im}\,\mathcal{H} = \mathrm{Si}(s) - \frac{\pi}{2} - s\,\mathrm{Ci}(s) + \sin s$, and $\mathcal{H}(-s) = \mathcal{H}(s)^*$. The bracket is the reflection tensor $\Pi^{ij}(r)\equiv\delta^{ij} - 2r^ir^j/r^2$:

$$\frac{d^3N_{\rm II.IV}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,x',y'}\mathcal{H}\big(k\cdot(y'-y)\big)\;n^i(x-y)\,\Pi^{ij}(y'-y)\,n^j(x'-y')\;\mathcal{D}^c(y',x')\,\mathcal{D}^c(y,x). \tag{4.15}$$

The kernel is symmetric under $(x,y)\leftrightarrow(x',y')$ while $s\to -s$; for commuting ρ only $\mathrm{Re}\,\mathcal{H}$ survives. $\mathcal{H}\simeq -\log|s|$ at $s\to 0$ (integrable).

5 Group III

5.1 Rows I and II

Both have the kernel $u^iv^jK^{ij}\log(B/A)$, the prefactor $-ig^4/8\pi^5k^+$ and the right factor R^{ab}. With $U^{c'd'}(y)U^{c'b}(z) = V^{bd'}$ the left factors add to

$$\frac{d^3N_{\rm III.I+III.II}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\,u^iv^jK^{ij}\log\frac{(x-z)^2}{(x-w)^2}$$
$$\times\Big\{U^{ab'}(x')V^{bd'}\{\rho^{b'}(x'),\rho^{d'}(y)\} - U^{ac'}(w')V^{be'}\rho^{c'}(x')\rho^{e'}(y) - U^{ab'}(x')\rho^b(y)\rho^{b'}(x')\Big\}R^{ab} + \text{c.c.} \tag{5.1}$$

5.2 Row III (first part) + Row V + Row VI

These have identical kernels, measures and phases; prefactors $ig^4/8\pi^4$, $ig^4/4\pi^4$, $ig^4/4\pi^4$. Let $\mathcal{D}^{c'}_q \equiv [U^{c'e'}(q) - U^{c'e'}(x')]\rho^{e'}(x')$ with $q=y'$ or z.

Step 1: colour. In units of $ig^4/8\pi^4$, Rows III(1) and VI have the left factor $f^{c'd'a}U^{d'b}(z)\mathcal{D}_q^{c'}$ and right factors that add to

$$U^{ac}(w)\{\rho^c(y),\rho^b(x)\} - U^{cb}(z)U^{ad}(y)U^{ce}(x)\{\rho^d(y),\rho^e(x)\} + 2\,U^{cb}(z)U^{ad}(y)U^{ce}(x)\rho^d(y)\rho^e(x) - 2\,U^{ad}(y)\rho^d(y)\rho^b(x)$$
$$= 2\big[U^{ac}(w)-U^{ac}(y)\big]\rho^c(y)\rho^b(x) - U^{ac}(w)\big[\rho^c(y),\rho^b(x)\big] + U^{cb}(z)U^{ad}(y)U^{ce}(x)\big[\rho^d(y),\rho^e(x)\big]. \tag{5.2}$$

Row V is $2f^{c'ca}\mathcal{D}_q^{c'}U^{ce}(x)[U^{ad}(y)-U^{ad}(w)]\rho^e(x)\rho^d(y) = -2f^{c'd'a}\mathcal{D}_q^{c'}U^{d'b}(x)[U^{ac}(w)-U^{ac}(y)]\rho^b(x)\rho^c(y)$. Adding, and using $U^{d'b}(z)U^{gb}(z)=\delta^{d'g}$:

$$\mathcal{C}_3(q) = 2\,f^{c'd'a}\,\mathcal{D}_q^{c'}\,\big[U^{d'b}(z)-U^{d'b}(x)\big]\rho^b(x)\,\big[U^{ac}(w)-U^{ac}(y)\big]\rho^c(y)$$
$$+\,f^{c'd'a}\,\mathcal{D}_q^{c'}\Big\{U^{ac}(w)U^{d'b}(z) - 2\,U^{ac}(y)U^{d'b}(z) + U^{ac}(y)U^{d'b}(x)\Big\}\big[\rho^c(y),\rho^b(x)\big]. \tag{5.3}$$

For commuting ρ only the first line remains: three LO factors $[U(\text{gluon})-U(\text{source})]\rho(\text{source})$ tied by one f, matching the three factors $n(x'-y')$, $n(x-z)$, $n(y-w)$ of the kernel.

Step 2: longitudinal integrals. With $\eta = p^+/k^+$, $s\equiv k\cdot(y'-z)$, $\vee\to\infty$ and (2.10):

$$\int_0^\infty \frac{d\eta}{1+\eta}\,e^{-i\eta s} = e^{is}E_1(is) \equiv J_0(s),$$
$$\int_0^\infty d\eta\left(\Big[\frac{1}{\eta(1+\eta)}\Big]_+ + \frac{1}{1+\eta}\right)e^{-i\eta s} = -\gamma_E - \log(is) = -\gamma_E - \log|s| - \tfrac{i\pi}{2}\,\mathrm{sgn}(s),$$
$$\int_0^\infty d\eta\,e^{-i\eta s} = \frac{-i}{s-i0}. \tag{5.4}$$

Step 3: result.

$$\frac{d^3N_{\text{III.III(1)+V+VI}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\Bigg\{$$
$$\Big[J_0(s)\;n(y'-z)\cdot n(x'-y')\;\;n(y-w)\cdot n(x-z)$$
$$+\big(\gamma_E+\log(is)\big)\;n(y'-z)\cdot n(x-z)\;\;n(y-w)\cdot n(x'-y')\Big]\mathcal{C}_3(y')$$
$$+\,\frac{i}{s-i0}\;n(y'-z)\cdot n(y-w)\;\Big[n(x-z)\cdot n(x'-y')\;\mathcal{C}_3(y') - n(x-z)\cdot n(x'-z)\;\mathcal{C}_3(z)\Big]\Bigg\} + \text{c.c.} \tag{5.5}$$

5.3 Row III, second part

Here $a=x-w$, $b=x-z$, $c=z-w$; the right colour factor is R^{ab} and the left one $\mathcal{L}_q^{ab}\equiv f^{c'd'a}\,U^{d'b}(z)\,\mathcal{D}_q^{c'}$. With $\eta=p^+/k^+$ the long bracket of the first term is

$$k^+\,[\dots]^{k'm} \equiv \mathfrak{B}^{k'm}(\eta) = \delta^{k'm}\left[\frac{e+\eta\,g}{1+\eta} - \phi + \frac{d_\perp\,\eta\,\phi}{(1+\eta)^2}\right] - \omega(\eta)\,E^{k'm} - G^{k'm} - E^{mk'},$$
$$\omega(\eta) \equiv \Big[\frac{1}{\eta(1+\eta)}\Big]_+ + \frac{1}{1+\eta}, \tag{5.6}$$

(the two $\delta^{k'm}$ brackets of the file are $g-\phi$ and $e-\phi$; the last bracket is $G^{k'm}+E^{mk'}$), and the short brackets of the second and third terms are both $G^{mk'}$. Therefore

$$\frac{d^3N_{\text{III.III(2)}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\int_0^\infty d\eta\;\frac{e^{-i\eta s}}{\eta B + A}\;n^m(y'-z)\Big\{n^{k'}(x'-y')\,\mathfrak{B}^{k'm}(\eta)\,\mathcal{L}_{y'}^{ab}$$
$$-\,\eta\,G^{mk'}\big[n^{k'}(x'-y')\,\mathcal{L}_{y'}^{ab} - n^{k'}(x'-z)\,\mathcal{L}_z^{ab}\big]\Big\}R^{ab}$$
$$+\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\;\frac{A}{B\,(B-A)}\;\log\frac{B}{A}\;\;n^m(w'-z)\,G^{mk'}\;n^{k'}(x'-z)\;\mathcal{L}_z^{ab}\,R^{ab}\;+\text{c.c.} \tag{5.7}$$

In the last term the p^+ integral is done: $\int_0^\infty d\eta\,A/[B(1+\eta)(\eta B+A)] = A\log(B/A)/[B(B-A)]$, regular at $B=A$, and $n^m(w'-z)G^{mk'}n^{k'}(x'-z) = \big[n(w'-z)\cdot\big(\frac{c}{c^2}-\frac{a}{2A}\big)\big]\big[b\cdot n(x'-z)\big]$. If wanted, $\int_0^\infty d\eta\,e^{-i\eta s}/(\eta B+A) = B^{-1}e^{isA/B}E_1(isA/B)$.

6 The $K\log$ sector across the groups

Six contributions have the kernel $u^i v^j K^{ij}\log[(x-z)^2/(x-w)^2]$, the same measure $\int_{w,w',x,y,z,x'}$ and phase $e^{-ik\cdot(w'-w)}$, and are of the form $X+\text{c.c.}$ In units of $\frac{1}{(2\pi)^3}\frac{ig^4}{8\pi^5 k^+}\,u^i v^j K^{ij}\log\frac{B}{A}$ their colour factors are

contribution	colour factor
I.II, Eq. (3.11)	$\frac{1}{2}\,\mathcal{D}_0^a\,\mathcal{S}^a$
I.III, Eq. (3.24)	$\mathcal{D}_0^a\,f^{abd}U^{de}(x)[U^{bc}(z)-U^{bc}(y)]\rho^e(x)\rho^c(y)$
I.IV	$\mathcal{D}_0^a\,V^{be}\rho^e(y)\,R^{ab}$
II.I ($1a$; $1b$ is the c.c.)	$-\frac{1}{2}\big[U^{ac'}(w')\{\rho^{c'}(x'),\rho^b(y)\}-U^{ad'}(x')V^{be'}\{\rho^{d'}(x'),\rho^{e'}(y)\}\big]R^{ab}$
III.I + III.II	$-\big\{U^{ab'}(x')V^{bd'}\{\rho^{b'}(x'),\rho^{d'}(y)\}-U^{ac'}(w')V^{be'}\rho^{c'}(x')\rho^{e'}(y)$ $-U^{ab'}(x')\rho^b(y)\rho^{b'}(x')\big\}R^{ab}$

Step 1: terms with V. With the shorthand $\rho'\equiv\rho(x')$, $\rho_y\equiv\rho(y)$ and colour indices suppressed, the terms containing V are

$$\begin{aligned}\text{I.IV}:&\quad \big[U(x')-U(w')\big]\rho'\,V\rho_y\,R,\\ \text{II.I}:&\quad +\tfrac{1}{2}\,U(x')\,V\{\rho',\rho_y\}\,R,\\ \text{III}:&\quad -U(x')\,V\{\rho',\rho_y\}\,R+U(w')\,V\rho'\rho_y\,R.\end{aligned}\tag{6.1}$$

The $U(w')V$ terms cancel; the $U(x')V$ terms leave $\frac{1}{2}\,U^{ab'}(x')V^{bd'}[\rho^{b'}(x'),\rho^{d'}(y)]R^{ab}$.

Step 2: terms without V from II and III.

$$-\tfrac{1}{2}\,U(w')\{\rho',\rho_y\}R+U(x')\rho_y\rho' R=\mathcal{D}_0^a\,\rho^b(y)\,R^{ab}-\Big(U^{ab'}(x')-\tfrac{1}{2}U^{ab'}(w')\Big)\big[\rho^{b'}(x'),\rho^b(y)\big]R^{ab}.\tag{6.2}$$

Step 3: add I.III. Its $U(z)$ term equals $-\mathcal{D}_0^a R_2^{ab}\rho^b(y)$ where $R_2^{ab}=-f^{acd}U^{cb}(z)U^{de}(x)\rho^e(x)$ is the second term of R^{ab}. With Step 2 the $U(z)$ dependence cancels up to a commutator, and the first five rows of the table add to $\Omega_0+\Omega_{\text{comm}}$,

$$\Omega_0=\mathcal{D}_0^a\Big\{f^{cbd}\,U^{ac}(w)\,\rho^b(y)\rho^d(x)-f^{abd}\,U^{bc}(y)\,U^{de}(x)\,\rho^e(x)\rho^c(y)\Big\},\tag{6.3}$$

$$\begin{aligned}\Omega_{\text{comm}}=&\tfrac{1}{2}\,U^{ab'}(x')\,V^{bd'}\big[\rho^{b'}(x'),\rho^{d'}(y)\big]R^{ab}-\Big(U^{ab'}(x')-\tfrac{1}{2}U^{ab'}(w')\Big)\big[\rho^{b'}(x'),\rho^b(y)\big]R^{ab}\\ &-\mathcal{D}_0^a\,f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\big[\rho^b(y),\rho^e(x)\big].\end{aligned}\tag{6.4}$$

Step 4: add I.II. Relabelling indices, $\Omega_0=-\mathcal{D}_0^a f^{hbc}U^{ah}(w)\rho^c(y)\rho^b(x)+\mathcal{D}_0^a f^{abc}U^{bd}(x)U^{ce}(y)\rho^d(x)\rho^e(y)$, so

$$\Omega_0+\tfrac{1}{2}\,\mathcal{D}_0^a\mathcal{S}^a=\tfrac{1}{2}\,\mathcal{D}_0^a\Big\{f^{hbc}\,U^{ah}(w)\big[\rho^b(x),\rho^c(y)\big]+f^{abc}\,U^{bd}(x)\,U^{ce}(y)\big[\rho^d(x),\rho^e(y)\big]\Big\}.\tag{6.5}$$

Result.

$$\boxed{\begin{aligned}&\frac{d^3N_{K\log}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\int_{w,w',x,y,z,x'}e^{-ik\cdot(w'-w)}\;u^iv^jK^{ij}\;\log\frac{(x-z)^2}{(x-w)^2}\;\Omega_{\text{tot}}+\text{c.c.},\\ &\Omega_{\text{tot}}=\Omega_{\text{comm}}+\tfrac{1}{2}\,\mathcal{D}_0^a\Big\{f^{hbc}\,U^{ah}(w)\big[\rho^b(x),\rho^c(y)\big]+f^{abc}\,U^{bd}(x)\,U^{ce}(y)\big[\rho^d(x),\rho^e(y)\big]\Big\}\end{aligned}}\tag{6.6}$$

Ω_{tot} contains only commutators of ρ. **For commuting ρ the six contributions cancel exactly.** If $[\rho^a(x),\rho^b(y)]=i\kappa f^{abc}\rho^c(x)\delta^2(x-y)$ with a convention-dependent κ, the last bracket becomes $i\kappa N_c\,[U^{ae}(w)+U^{ae}(x)]\rho^e(x)\,\delta^2(x-y)$ (using $f^{hbc}f^{bce}=N_c\delta^{he}$ and $f^{abc}U^{bd}U^{ce}=U^{ah}f^{hde}$), i.e. a local term with one ρ less.

7 The negative logarithms and reality

- In Group I Row III the logarithms of the sign-indefinite combination $(x-w)^2-(x-z)^2$ cancel completely, with all poles $1/[(x-w)^2-(x-z)^2]^{1,2}$.
- What remains is the constant in $\log[-B/A]=\log[B/A]+i\sigma\pi$. It does not cancel inside Row III, and no other row of the file contains a compensating term. With the prefactor i it gives a real kernel:

$$\left.\frac{d^3N_{\text{I.III}}}{d^3k}\right|_{\pi}=-\,\sigma\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\;u^i\,v^j\,K^{ij}\;\mathcal{W}_{\text{III}}\;+\;\text{c.c.}\tag{7.1}$$

- Every contribution of Groups I and III is $X + \text{c.c.}$ and therefore real whatever the phase of X. For commuting ρ, where the colour factors are real, an $i\times$(real kernel) term gives $\sin[k\cdot(w'-w)]$ and the π term gives $\cos[k\cdot(w'-w)]$:

$$\frac{d^3N_{\text{I.III}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^5}\,\frac{1}{k^+}\int u^i v^j\,\mathcal{W}_{\text{III}}\Big\{\sin\big[k\cdot(w'-w)\big]\,\mathcal{R}^{ij}_{\text{real}} - \sigma\pi\,\cos\big[k\cdot(w'-w)\big]\,K^{ij}\Big\}, \tag{7.2}$$

  with $\mathcal{R}^{ij}_{\text{real}} = K^{ij}\log[B/A] - \frac{1}{2}S^{ij}\log[c^2/A] + \mathcal{A}_0 T^{ij}$.

- Group I Row II has no explicit i, but its kernel $\mathcal{R}^i$ is real and odd under $(q,p)\to(-q,-p)$, so its Fourier transform is $i\times$real, consistent with (3.11).

- Group II: $N_{1b} = (N_{1a})^*$; the second part is the square (4.11); Row IV has a real symmetric tensor and $\mathcal{H}(-s) = \mathcal{H}(s)^*$. All real, all logarithms positive.

8 The simplified finite term

After the reductions the finite term is the sum of the following pieces.

	piece	equation
1	K log sector: I.II, I.IV, II.I first part, III.I, III.II and the $K\log(B/A)$ part of I.III	(6.6) (commutators only)
2	rest of I.III: $i\sigma\pi K^{ij} - \frac{1}{2}S^{ij}\log\frac{c^2}{A} + \mathcal{A}_0 T^{ij}$, colour $\mathcal{W}_{\text{III}}$	(3.13), (3.24)
3	I.V first part	(3.27)
4	I.V second part ($N_{2a} + N_{2b\,\text{rest}}$)	(3.30)
5	II.I second part ($2a$–$2f$), perfect square	(4.11), (4.12)
6	II.IV	(4.15)
7	III.III first part + III.V + III.VI	(5.5)
8	III.III second part	(5.7)

For commuting ρ, piece 1 vanishes and the colour factor of piece 7 is the first line of (5.3).

9 Assumptions and points to check

1. **Behaviour of I.III at $w\to z$.** $\mathcal{W}_{\text{III}}$ does not depend on w, and for $c = z-w\to 0$

$$\mathcal{R}^{ij}_{\text{III}} \;\longrightarrow\; -\frac{\delta^{ij}}{2\,(z-w)^2}\left\{\log\frac{(z-w)^2}{(x-w)^2} - i\sigma\pi\right\} + \mathcal{O}\!\left(\frac{1}{|z-w|}\right), \tag{9.1}$$

   since $K^{ij}\to\delta^{ij}/2c^2$, $S^{ij}\to\delta^{ij}/c^2$, $K^{ij}\log[B/A] = \mathcal{O}(1/|c|)$, $\mathcal{A}_0T^{ij} = \mathcal{O}(1/|c|)$. $\int d^2w\,\log(c^2)/c^2$ does not converge at $w\to z$ and the phase does not help. The source is the last term of N_{3b} and the $i\sigma\pi$ term. Piece 2 of Section 8 is therefore not finite by itself: either a subtraction is missing from the file, or the $\Lambda\to 0$ limit does not commute with the w integral there.

2. **Sign σ.** Fixed by the $i\epsilon$ of the energy denominators. The reduction assumes all "-1" inside the logarithms are continued in the same way; otherwise extra multiples of $2\pi i\,M^{ij}$ appear, and those have poles at $A = B$.

3. **Row II.** I renamed the integration momentum and assumed the x' integral. The equality (3.11) is exact (operator level) because $\mathcal{S}^a$ is exactly antisymmetric; the dropped term $\mathcal{E}^i$ is individually not integrable at $l\to 0$ and at $p\to 0$, so the statement is that it cancels in the antisymmetric combination.

4. **Plus prescriptions.** The closed-form longitudinal integrals assume (2.10). A different convention changes only those closed forms, not the brackets $\mathfrak{B}$, $\mathfrak{B}_V$.

5. **N_{2a} of I.V.** The merging with $N_{2b\,\text{rest}}$ uses the limits $\int_{k^+}^0 dq^+$ as written. If $\int_0^{k^+}$ is meant, the $\bar{E}^{im}$ term of N_{2a} has the opposite sign, the kernels differ by that term, and the $U(x)$ cancellation is incomplete.

6. **Distributions.** $\int_0^\infty d\eta\, e^{-i\eta s} = -i/(s-i0)$ is the $\vee\to\infty$ limit in the distributional sense and includes a $\pi\delta(k\cdot(y'-z))$ piece.

7. **Commutators.** Ω_{tot} and the second line of (5.3) are left as commutators; evaluating them needs the normalisation of $[\rho,\rho]$.

8. **Convergence** of the transverse integrals in pieces 3–8 was not examined.

A Fourier transform of $K^{ij}\log(b^2/a^2)$

Conventions: $\tilde f(k)=\int d^2r\, e^{ik\cdot r}f(r)$, $\lambda(k)\equiv\log(4/k^2)-2\gamma_E$.

$$\widetilde{n^i}(k)=\frac{2\pi i\,k^i}{k^2},\qquad \widetilde{n^i\log r^2}(k)=\frac{2\pi i\,k^i}{k^2}\,\lambda(k),\qquad \widetilde{\log r^2}(k)=-\frac{4\pi}{k^2}\ \text{(regularised)},\qquad \widetilde{r^{-2}}(k)=\pi\lambda(k)+\text{const.} \tag{A.1}$$

For a product $f(a)\,g(b)\,h(a-b)$:

$$\int d^2a\,d^2b\,e^{iq\cdot a+il\cdot b}f(a)g(b)h(a-b)=\int\frac{d^2k}{(2\pi)^2}\,\tilde f(q-k)\,\tilde g(l+k)\,\tilde h(k). \tag{A.2}$$

Write $K^{ij}\log\frac{b^2}{a^2}=T_1+T_2+T_3+T_4$ with $T_1=n^i(c)n^j(b)\log\frac{b^2}{a^2}$, $T_2=-n^i(a)n^j(c)\log\frac{b^2}{a^2}$, $T_3=-n^i(a)n^j(b)\log\frac{b^2}{a^2}$, $T_4=\frac{\delta^{ij}}{2c^2}\log\frac{b^2}{a^2}$, and $p\equiv q+l$.

- T_3 and T_4 factorise: $\tilde T_3=4\pi^2\frac{q^il^j}{q^2l^2}\log\frac{q^2}{l^2}$, $\tilde T_4=\frac{2\pi^2\delta^{ij}}{p^2}\log\frac{q^2}{l^2}$.
- The $\log b^2$ part of T_1 and the $\log a^2$ part of T_2 factorise: $-4\pi^2\frac{q^ip^j}{q^2p^2}\lambda(p)$ and $+4\pi^2\frac{p^il^j}{p^2l^2}\lambda(p)$.
- The remaining parts are one-loop integrals, $\int d^2k\,\widetilde{\log r^2}(k-q)\,\varphi_1(k)$ with $\varphi_1=\frac{k^i(k+l)^j}{k^2(k+l)^2}$, and $\int d^2k\,\widetilde{\log r^2}(k+l)\,\varphi_2(k)$ with $\varphi_2=\frac{(q-k)^ik^j}{(q-k)^2k^2}$, where

  $$\int d^2k\,\widetilde{\log r^2}(k-q)\,\varphi(k)=-4\pi\int d^2k\,\frac{\varphi(k)-\varphi(q)\,\theta(1-|k-q|)}{(k-q)^2}+4\pi^2(\log4-2\gamma_E)\,\varphi(q). \tag{A.3}$$

  The constants $\log4-2\gamma_E$ cancel against those in $\lambda(p)$.
- After contraction with l^j the loop integrals are done with complex coordinates ($k=k_x+ik_y$, $n(k)\leftrightarrow1/\bar k$, $n(k)\cdot l=\frac12[l/k+\bar l/\bar k]$): the integrand is a product of a function with simple poles in k and one with simple poles in $\bar k$, and after partial fractions only $\int d^2k\,\frac{1}{(k-\alpha)(\bar k-\bar\beta)}=\pi\log\frac{\Lambda^2}{|\alpha-\beta|^2}$ is needed (the cutoff Λ cancels).

The sum is

$$l^j\int d^2a\,d^2b\,e^{iq\cdot a+il\cdot b}K^{ij}\log\frac{b^2}{a^2}=2\pi^2\log\frac{q^2}{l^2}\left[\frac{q^i}{q^2}-\frac{p^i}{p^2}-\frac{l^2\,q^i}{p^2q^2}\right]=-2\pi^2\,l^2\,\mathcal{R}^i(q,p). \tag{A.4}$$

The partial-fraction sum was evaluated numerically and identified with this closed form (agreement 10^{-11} at 300 random momenta; scale invariance checked; the partial-fraction formula for the regulated loop integral was checked against a direct two-dimensional numerical integration).

B Numerical checks

statement	check
(3.24) vs. the 22 original terms of I.III	2000 random points, complex logs, $<10^{-8}$
$\mathcal{A}_0=\theta_w/\Delta$; Row IV bracket $=K^{ij}$	same run
$R^i/(p^2q^2)=\mathcal{R}^i+2\mathcal{E}^i$	random momenta, 10^{-16}
(3.8)	Appendix A
five colour structures $=\Omega_0+\Omega_{\rm comm}$; $\Omega_0+\frac12\mathcal{D}_0\mathcal{S}$	$SU(2)$, non-commuting matrix-valued ρ, 10^{-13}
$\mathcal{C}_3(q)$, Eq. (5.3)	$SU(2)$, non-commuting ρ, 10^{-14}
colour bracket of II.I $=R'R$	$SU(2)$
(4.12) and (4.11) vs. original $2a$–$2f$	3000 points, $d_\perp=2$ and $\neq2$, $<10^{-10}$; η integral done numerically
brackets $\mathfrak{B}$ (5.6) and $\mathfrak{B}_V$ (3.29) vs. original	500 points each, 10^{-14}
longitudinal integrals (h_0, Ein, J_0, $\gamma_E+\log(is)$, $\mathcal{H}$)	numerical quadrature, 10 digits